New impetus to materials research - democratizing a frontier research tool

Reporting

Project Information

LynXes

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Coordinated by
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Hungary

Periodic Reporting for period 2 - LynXes (New impetus to materials research - democratizing a frontier research tool)

Reporting period: 2025-02-01 to 2026-01-31

Summary of the context and overall objectives of the project

The discovery of X-rays and their interaction with matter more than a century ago revolutionized many fields in research and everyday technologies. X-rays can be used to image nonvisible objects like bones in living organisms, cracks in metallic welds, etc., due to their low and matter dependent absorption. However, a better control of the X-ray – matter interaction gives us more sophisticated tools to analyze unknown material. Wealthy business sectors are built upon such well-established X-ray based analytical methods like X-ray diffraction (XRD) or X-ray fluorescence spectroscopy (XRF), which provide detailed information on the atomic structure and elemental composition, respectively. The market size of these methods reach €700M/year and €1.5B/year respectively, including the Pharma, Biotech, Chemical, Waste Recycling, Mining, Food, etc. industries. In this line, high-energy-resolution X-ray spectroscopies (HERXS) provide the most detail-rich way to examine physical and chemical properties of materials and have been used for more than three decades by scientists all over the world.
However, the market for using HERXS was non-existent until the very recent years, as these frontier analytical tools were considered to be applicable only at costly large-scale X-ray source facilities (like €1B cost synchrotrons or X-ray free electron lasers) with limited and restricted access. The usage of these methods with lab-size X-ray tubes were thought to be unfeasible.

A prominent application is the development and utilization of novel catalyst materials fulfilling the ever more important requirements of sustainability, environment-friendliness, ethical and economical adequacy. Lab-scale HERXS is an excellent option to expand their analytical toolbox on developing e.g. novel Pharma-related active agents with new, green and economic catalysts, follow reaction routes and understand both formulation mechanisms of the active molecules as well as their mode of action.

LynXes has created desktop-sized HERXS devices that provide detailed analytical results comparable to billion-euros-cost large-scale facilities but at a fraction of the cost. This breakthrough allows wider accessibility to HERXS for researchers, scientists, and industrial development teams of various sectors such as green chemistry, pharma, waste management, data storage, but most of all, in producing and storing green energy. LynXes has diverse use cases spanning industrial R&D, academia, and in the long-term, quality control in production. It enables companies to strengthen their R&D potential, improve product development, and achieve cost savings in product quality control. Universities benefit from on-site access to HERXS techniques, enabling them to educate students, establish productive academic-industrial collaboration, and contribute to large-scale facilities' utilization. Although this also creates a good business opportunity for us, the main benefit will be the early engagement to LynXes technology.

Work performed from the beginning of the project to the end of the period covered by the report and main results achieved so far

The concept was formulated in 2014 as an academic project (under ERC). The aim was to create a demonstration system which would enable us to use large-scale facilities more effectively (practice & optimize the setup for measurements). During the development, we found that the setup works as a

stand-alone system, so a proof of concept was created in 2016 and the initial results were published. Since then we have been using and optimizing the technology for our own scientific use and validated it in the lab and later in a relevant environment in collaboration with international research partners in 2021. In parallel with the technological development we presented our prototype and results at various conferences and received great interest both from industry and academia. This led us to create LynXes Ltd, a start-up company to commercialize this breakthrough technology through validated industrial use cases.

In the first year of our EIC Accelerator project, we advanced our instrument portfolio from TRL5 to TRL6. This involved a complete redesign of the spectrometer components, preparing them for testing and validation by external users, showcasing their capabilities to a broader audience of potential customers. Simultaneously, we launched our branding efforts and began cultivating a new market by actively communicating and promoting our results and the benefits of our technology.

In the second year we have finished product development, established our comapny structure, started to create the market, and upgraded our Business Plan.

Progress beyond the state of the art and expected potential impact (including the socio-economic impact and the wider societal implications of the project so far)

Table-top HERXS spectrometers as generic analytical tools open an entire new horizon in industrial and academic R&D. Our target market (using HERXS in industrial R&D) theoretically already exists, but it is so narrow that it doesn't really show, due to the current very high price and the very limited accessibility of the technology at large-scale facilities. Our exploitation strategy thus is based on not just technology but knowledge transfer as well, because we have to demonstrate (e.g. face-to-face events, pilot possibilities, symposia, releases in relevant sectoral media, etc.) for our potential clients how they can best exploit the technology we offer.

As our innovation will take relevant R&D processes to a new level, our product will be not only a market but also a technological benchmark for future products and technologies. This will result in innovation pressures for our potential competitors. Furthermore, our solution will give the opportunity to develop better products, which will positively affect the growth of relevant sectors as well. Finally, the most advanced competing material analytic developments currently are in the US, China, and Japan, while LynXes is a strong European player. So, our project will put Europe at the forefront of the world, and it will strengthen European technological and industrial sovereignty.

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